Commodity Options & Derivatives Analytics Platform

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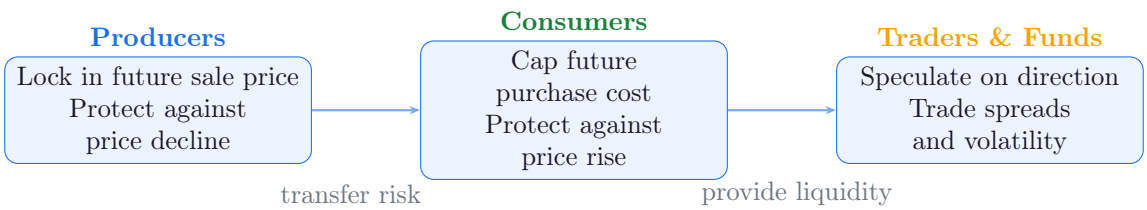
1 Introduction

1.1 What is a Commodity Derivative?

A commodity derivative is a financial contract whose payoff depends on the future price of a raw material: crude oil, natural gas, gold, corn, and so on. Unlike equity derivatives, commodity derivatives are almost universally priced on **futures prices**, not spot prices, because:

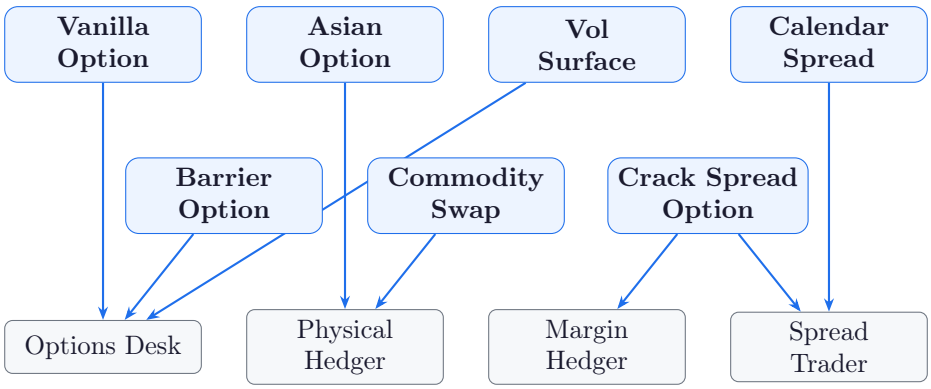
- Futures are liquid, transparent and observable at every maturity on organised exchanges.
- The futures price already embeds storage costs, financing and convenience yield. No additional cost-of-carry adjustment is needed in the pricing model.
- Physical delivery is impractical for most market participants; futures allow pure price exposure without storage obligations.

Three types of participants use commodity derivatives:



1.2 Product Overview

CODAP implements six pricing engines covering the full spectrum of commodity derivatives used by trading desks:

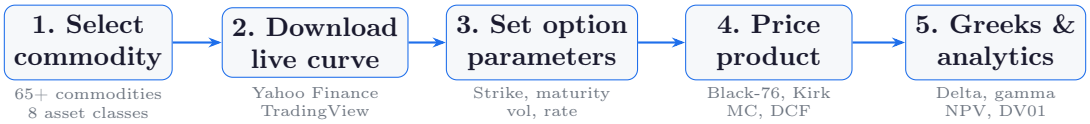


Tab	Product	Model
Vanilla Options	European call/put on futures	Black-76
Asian Options	Arithmetic/geometric average-price option	MC + Kemna-Vorst
Crack Spread	Option on refining margin $F_2 - F_1$	Kirk (1995)
Calendar Spread	Option on curve slope $F_{\text{near}} - F_{\text{far}}$	Kirk on term structure
Commodity Swaps	Fixed-for-floating commodity swap	Discounted cashflows
Barrier Options	Knock-in / knock-out (daily monitoring)	Daily Monte Carlo
Vol Surface	Parametric implied volatility surface	Black-76 + skew
Forward Curve	Live futures curve + cost-of-carry extension	Yahoo / TradingView

2 Architecture

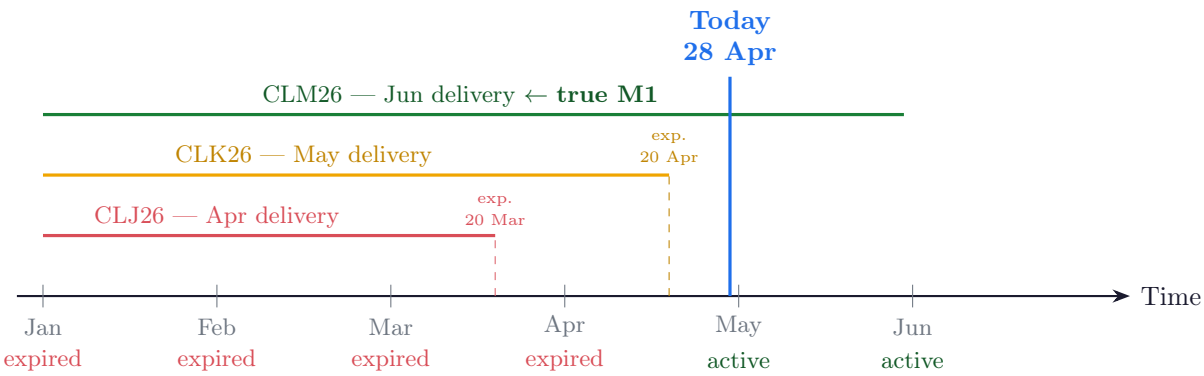
2.1 The Pricing Pipeline

Every product in CODAP follows the same five-step pipeline:



2.2 Expiry-Aware Ticker Generation

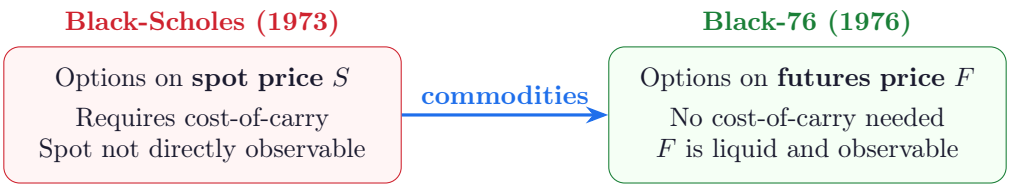
A critical design choice: CODAP always downloads the **true front month**, not an expired contract. Standard futures expire approximately on the 20th of the month before delivery.



Without this fix, Yahoo Finance returns stale prices from expired contracts: on April 28 2026 it would return the last traded price of CLJ26 (expired March 20) instead of the live price of CLM26 (the true front month).

3 Black-76 Vanilla Option Pricer

3.1 Why Black-76 and Not Black-Scholes?



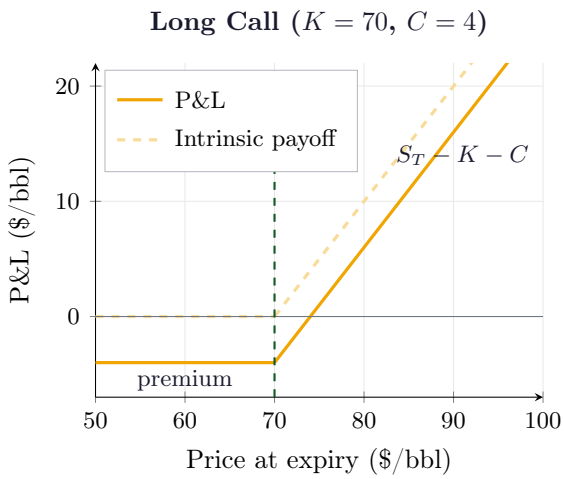
The key insight: under the *futures measure*, the futures price F has zero drift. The discounting e^{-rT} accounts for the fact that option premium is paid upfront but the futures payoff occurs at expiry.

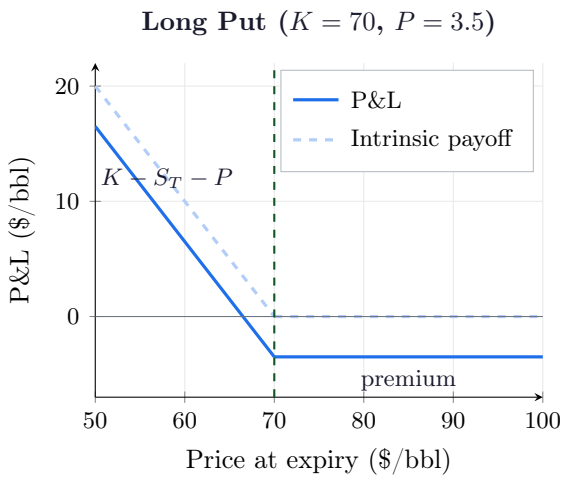
3.2 Pricing Formulas

$$\boxed{C = e^{-rT}\left[FN(d_1) - KN(d_2)\right]} \qquad \boxed{P = e^{-rT}\left[KN(-d_2) - FN(-d_1)\right]} \tag{1}$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{2}$$

3.3 Payoff Diagrams



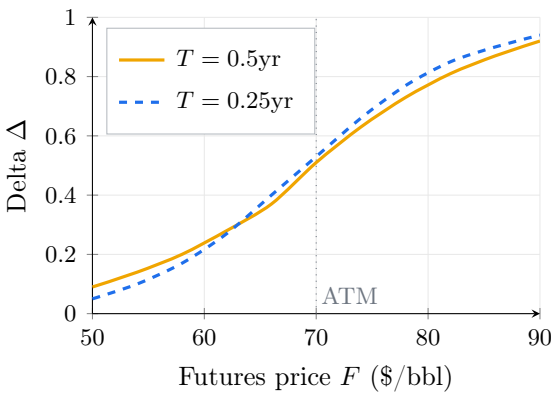


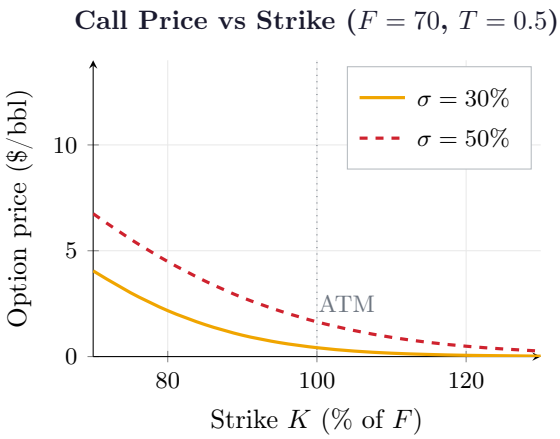
3.4 Greeks

Greek	Formula (call)	Economic interpretation
Δ	$e^{-rT}N(d_1)$	*Delta*: number of futures needed to delta-hedge one option. ATM call ≈ 0.50. Deep ITM $\to 1$, deep OTM $\to 0$.
Γ	$\dfrac{e^{-rT}N'(d_1)}{F\sigma\sqrt{T}}$	*Gamma*: rate of change of delta per unit move in F. High Γ means the hedge needs frequent and costly rebalancing.
$\mathcal{V}$	$\dfrac{e^{-rT}FN'(d_1)\sqrt{T}}{100}$	*Vega*: P&L per 1% increase in implied vol. Long options have positive vega — they gain when vol rises.
Θ	$\dfrac{-e^{-rT}FN'(d_1)\sigma}{2\sqrt{T}\cdot 365}$	*Theta*: daily time decay. Always negative for long options. Largest near expiry when the option is at-the-money.
ϱ	$\dfrac{-Te^{-rT}[FN(d_1)-KN(d_2)]}{100}$	*Rho*: P&L per 1% change in risk-free rate. Small for short-dated commodity options; much less important than vega.

3.5 Delta-Gamma Sensitivity Profile

Delta vs Futures Price ($K = 70$, $\sigma = 30\%$)





3.6 Put-Call Parity

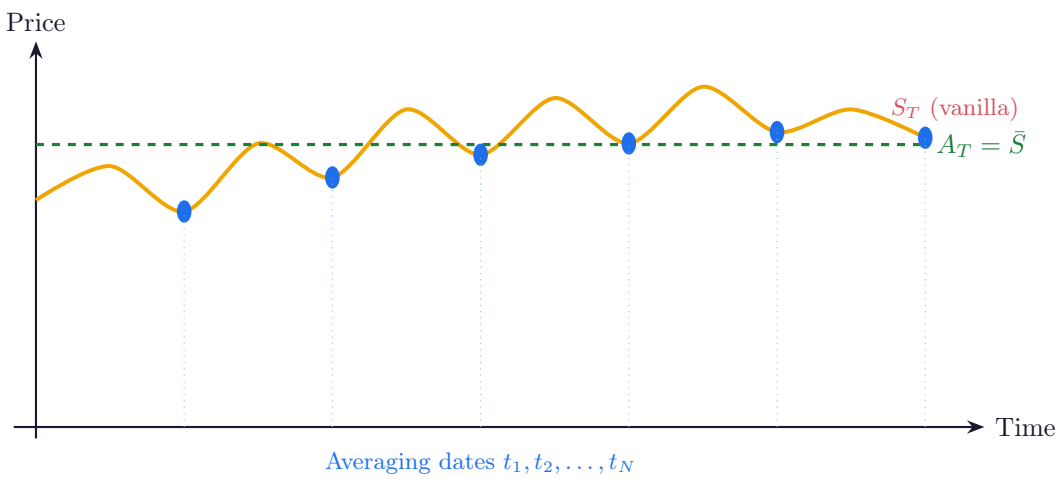
$$C - P = e^{-rT}(F - K) \tag{3}$$

CODAP verifies this for every option pair. The residual is always below 10^{-4} and displayed as ✓ `verified` in the dashboard.

4 Asian Option Pricer

4.1 Why Asian Options in Commodity Markets?

Asian options pay off on the **average price** over the option's life, not the final price. This is the natural instrument for physical commodity participants whose costs and revenues are priced on monthly averages.



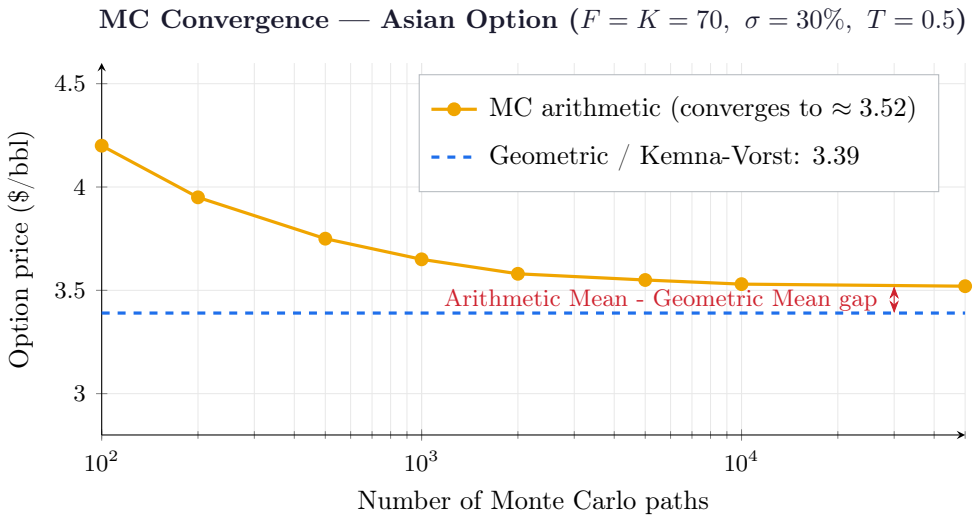
Why averaging reduces the option premium: Averaging smooths out price spikes and collapses. The effective volatility of the average is lower than that of the spot: $\sigma_{\text{avg}} < \sigma_{\text{spot}}$. A cheaper premium makes Asian options the preferred hedging instrument for buyers who must anyway purchase at the monthly average.

4.2 Monte Carlo Pricing

$$S_{t+\Delta t} = S_t \exp\left(-\tfrac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t}\, Z\right), \quad Z \sim \mathcal{N}(0,1) \tag{4}$$

$$C_{\text{Asian}} = e^{-rT} \cdot \frac{1}{M} \sum_{j=1}^{M} \max\left(\frac{1}{N} \sum_{i=1}^{N} S_{t_i}^{(j)} - K,\ 0 \right) \tag{5}$$

4.3 MC Convergence and Kemna-Vorst Benchmark



AM-GM ordering: The arithmetic average $\geq$ geometric average (AM-GM inequality). Therefore the MC arithmetic price is always $\geq$ the analytical Kemna-Vorst price. CODAP verifies this automatically as a numerical sanity check.

4.4 Geometric Average — Kemna-Vorst (1990)

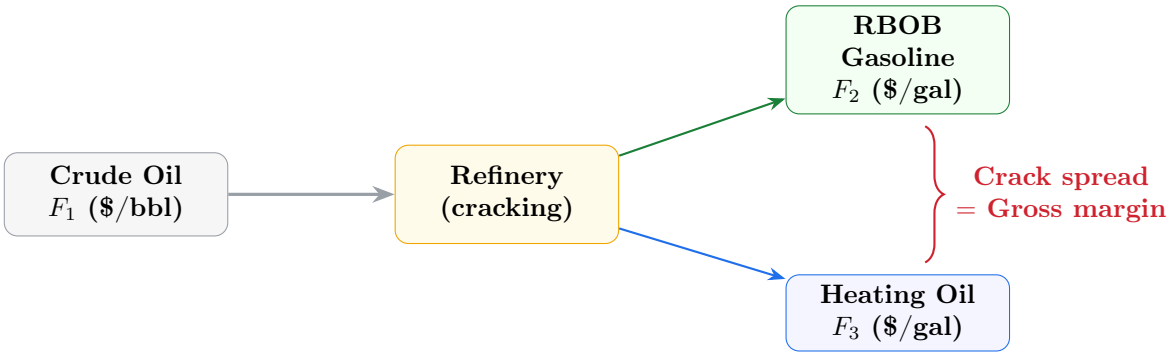
$$\sigma_g = \sigma \sqrt{\frac{2N+1}{6(N+1)}}, \qquad b_g = \frac{1}{2}\left(-\frac{\sigma^2}{6} + \sigma_g^2 \right), \qquad F_g = F \cdot e^{b_g T} \tag{6}$$

The geometric Asian is then priced as a Black-76 option with parameters (σ_g, F_g).

5 Crack Spread Option Pricer

5.1 The Refining Margin

A refiner is **naturally long the crack spread**: it buys crude oil (F_1) and sells refined products (F_2, F_3). It hedges by buying a **put on the crack spread** — protecting against margin compression.



The most common crack spread configurations:

Name	Formula	Typical value
3-2-1 crack	$\frac{1}{3}(2 \cdot F_{\text{RBOB}} \cdot 42 + F_{\text{HO}} \cdot 42) - F_{\text{WTI}}$	\$12–20/bbl
Simple crack	$F_{\text{RBOB}} \cdot 42 - F_{\text{WTI}}$	\$10–16/bbl
Jet crack	$F_{\text{Jet CIF NWE}} - F_{\text{Brent}}$	\$70–100/mt
Spark spread	$F_{\text{power}} - \text{HR} \times F_{\text{gas}}$	\$3–8/MWh

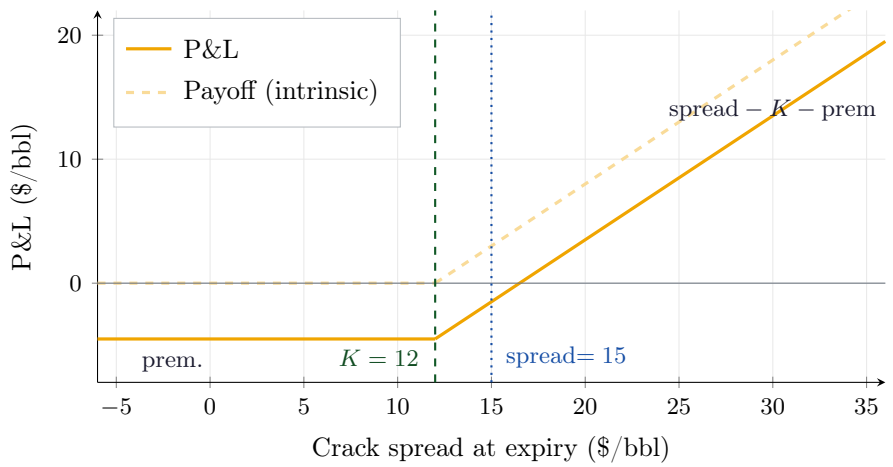
5.2 Kirk (1995) Approximation

$$\alpha = \frac{F_2}{F_2 + Ke^{-rT}}, \qquad \boxed{\sigma_{\text{Kirk}} = \sqrt{\sigma_1^2 + (\alpha\sigma_2)^2 - 2\rho\,\sigma_1\sigma_2\,\alpha}} \tag{7}$$

Why Kirk and not Margrabe?

Margrabe's exact formula only prices exchange options with $K = 0$. Crack spread options always have $K > 0$ (the target margin). Kirk (1995) extends Margrabe to non-zero strikes and is the industry standard. Accuracy is excellent for $\rho > 0.70$, which always holds for crude vs refined products (typical $\rho \approx 0.85$–0.95).

Crack Spread Call — P&L at Expiry ($K = 12$, premium $= 4.5$)

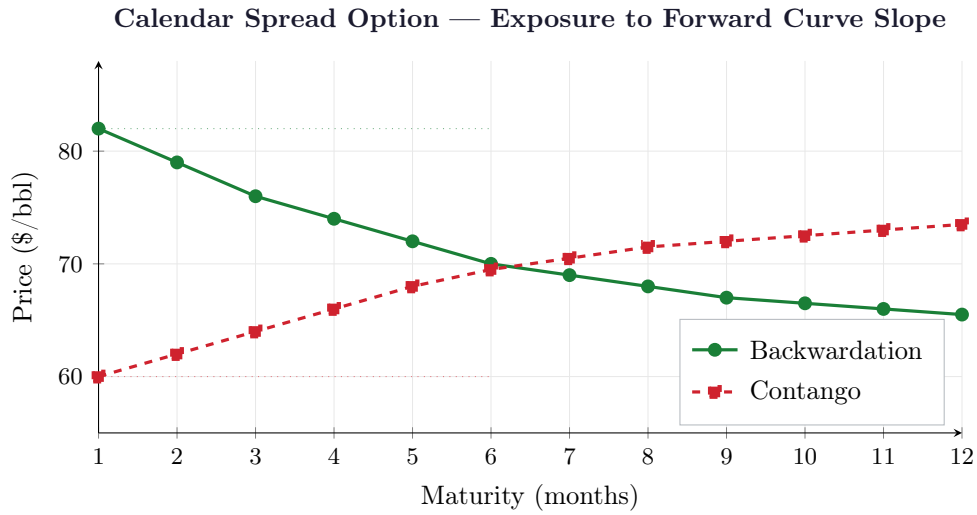


6 Calendar Spread Option Pricer

6.1 What a Calendar Spread Option Trades

A calendar spread option gives exposure to the **slope of the forward curve**: the difference between two delivery dates on the same commodity.

- **Call on M1-M6 spread**: pays off when backwardation widens (M1 rises relative to M6). Used by traders who believe near-term supply will tighten further.
- **Put on M1-M6 spread**: pays off when contango deepens. Used by storage players who want to hedge their carry trade.



6.2 Kirk on the Term Structure

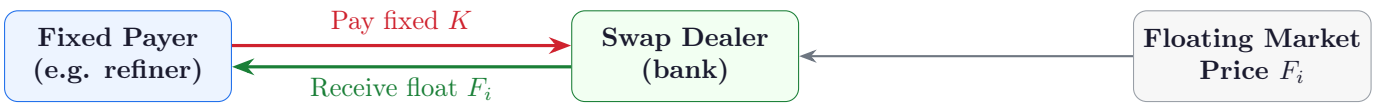
$$\alpha = \frac{F_{\text{near}}}{F_{\text{near}} + Ke^{-rT}}, \qquad \sigma_{\text{Kirk}} = \sqrt{\sigma_{\text{far}}^2 + (\alpha\,\sigma_{\text{near}})^2 - 2\rho\,\sigma_{\text{far}}\,\sigma_{\text{near}}\,\alpha} \tag{8}$$

CODAP defaults to $\sigma_{\text{near}} = 1.05\,\sigma$ and $\sigma_{\text{far}} = 0.90\,\sigma$, reflecting the empirical observation that short-dated contracts are more volatile than deferred contracts.

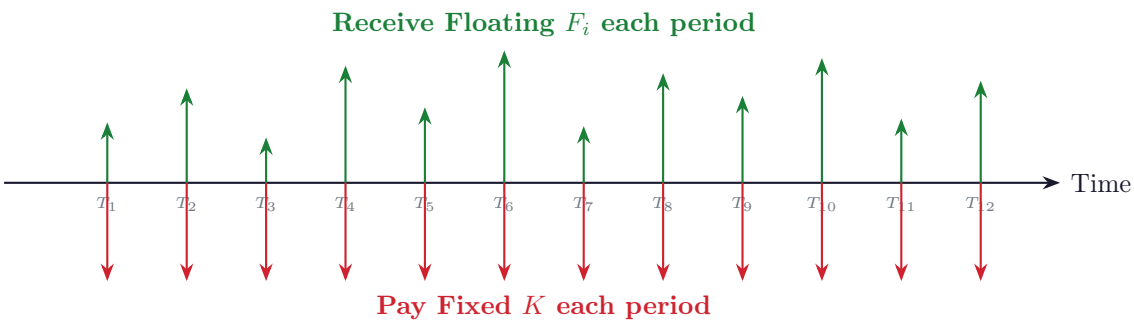
7 Commodity Swap Pricer

7.1 What is a Commodity Swap?

A commodity swap is a bilateral contract in which one party pays a **fixed price** K per period and receives the **floating market price** F_i.



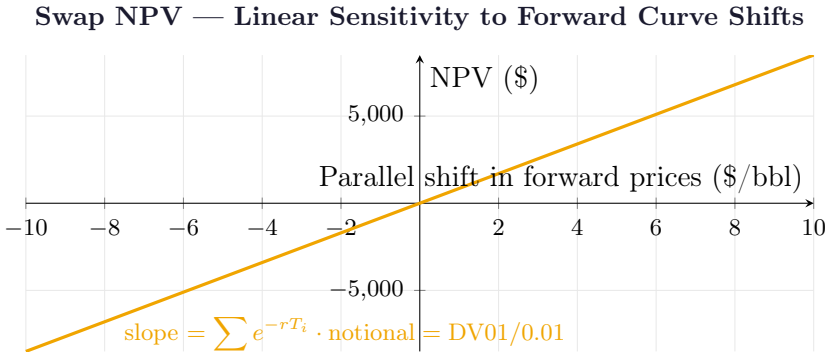
7.2 Cash Flow Diagram



$$\boxed{\text{NPV} = \sum_{i=1}^{N} e^{-rT_i}(F_i - K)\cdot\text{notional}} \qquad K^* = \frac{\sum_{i=1}^{N} e^{-rT_i} F_i}{\sum_{i=1}^{N} e^{-rT_i}} \tag{9}$$

K^* is the **fair fixed rate**: the level at which NPV $= 0$ at inception. It is the present-value-weighted average of all forward prices.

7.3 DV01 and Sensitivity



DV01 (Dollar Value of 1 basis point) measures the change in NPV for a 0.01% parallel shift in all forward prices — the primary sensitivity metric for swap portfolios.

8 Barrier Option Pricer

8.1 Types of Barrier Options

A barrier option activates (knock-in) or terminates (knock-out) if the underlying touches a barrier level B during the option's life.

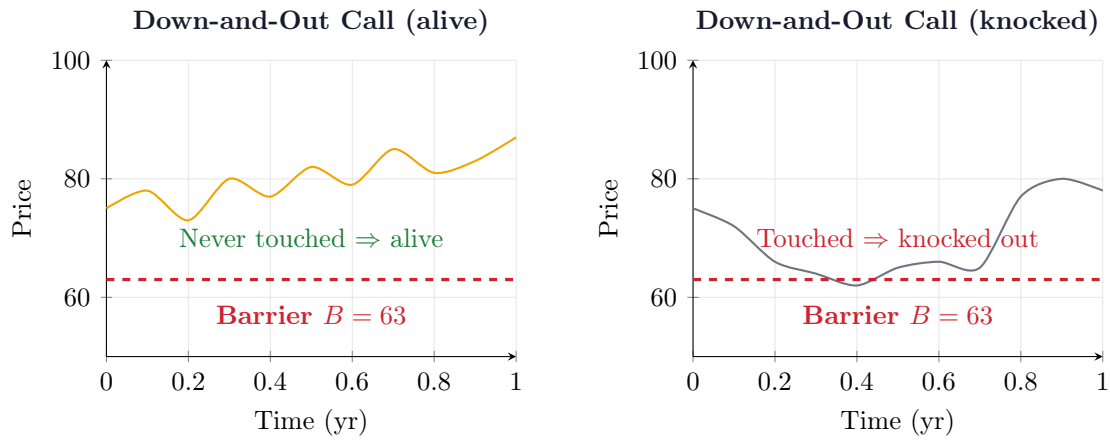
Type	Barrier	Payoff condition	Primary use
Down-and-Out (KO)	$B < S$	Vanilla payoff if $\min_t S_t > B$	Most common in energy
Down-and-In (KI)	$B < S$	Vanilla payoff only if $\min_t S_t \leq B$	Speculative
Up-and-Out (KO)	$B > S$	Vanilla payoff if $\max_t S_t < B$	Producer hedge
Up-and-In (KI)	$B > S$	Vanilla payoff only if $\max_t S_t \geq B$	Structured products

Why Down-and-Out Call is most common in energy: A consumer buys a call to hedge against rising prices. Adding a down barrier makes the option cheaper — if the price collapses below B, the option is knocked out, but the consumer doesn't need protection anyway because the market is giving them a cheap purchase price.

8.2 Daily Monte Carlo Simulation

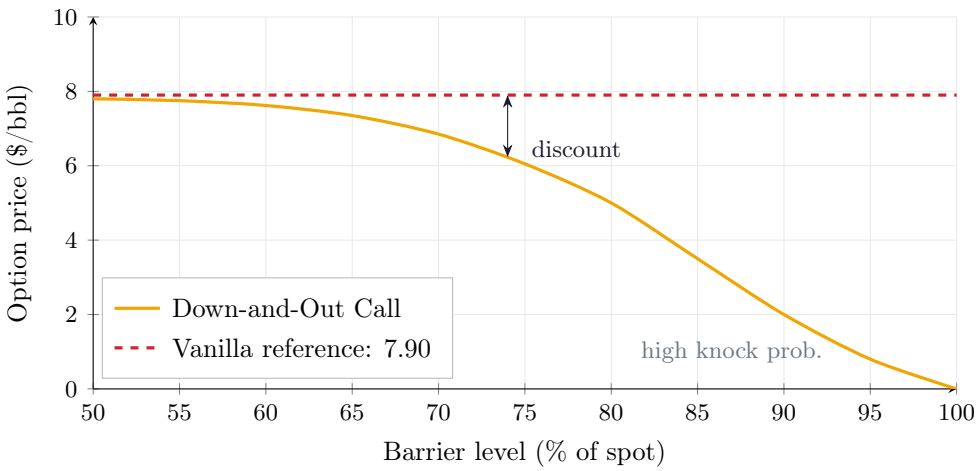
Analytical barrier formulas assume a continuous barrier. In practice, barriers are monitored **daily** (at settlement), so daily-step MC is the correct method:

$$C_{\text{KO}} = e^{-rT} \cdot \mathbb{E}\Big[\max(S_T - K, 0) \cdot \mathbf{1}_{\{\min_{0 \leq t \leq T} S_t > B\}}\Big] \tag{10}$$



8.3 Price vs Barrier Level

Down-and-Out Call Price vs. Barrier Level ($F = K = 70,\ \sigma = 30\%,\ T = 0.5$)



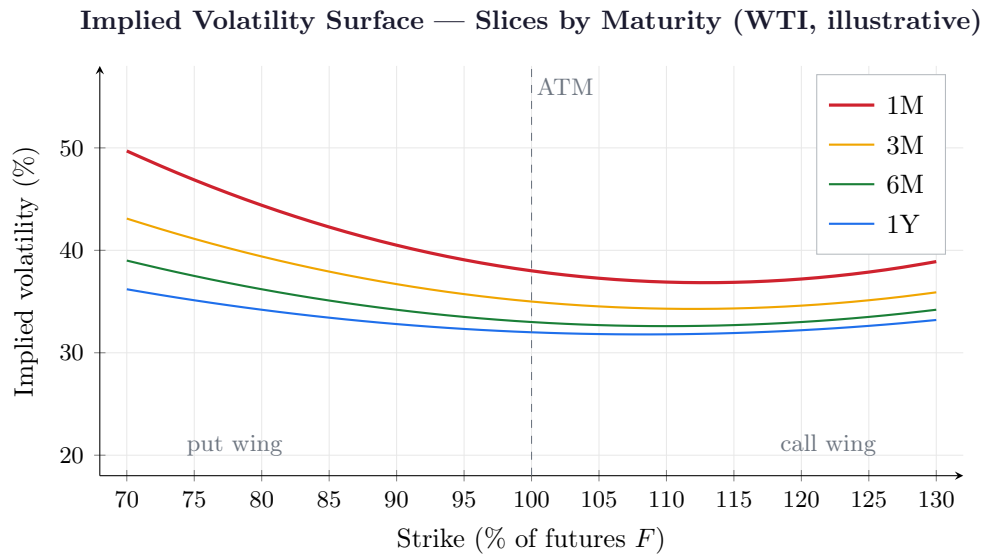
As the barrier moves closer to the spot ($B \to S$), the knock-out probability increases and the barrier price converges to zero. At $B = 0$, the barrier option equals the vanilla. The **discount** vs vanilla is the price the buyer pays for the knock-out risk.

9 Volatility Surface

9.1 The Smile and Skew

In real commodity markets, implied volatility is not flat across strikes. Two empirical patterns emerge:

- **Skew** ($\partial\sigma/\partial K < 0$): OTM puts more expensive than OTM calls. Reflects fear of sharp downside moves (supply shocks, demand collapse, geopolitical events).
- **Smile** (curvature): both wings more expensive than ATM. Reflects fat tails in the price distribution — extreme moves in both directions are priced in.



9.2 Parametric Model

$$\sigma(K,T) = \underbrace{\sigma_{\text{ATM}}(1+\nu\sqrt{T})}_{\text{term structure}} + \underbrace{\text{skew}\cdot\ln\tfrac{K}{F}}_{\text{slope}} + \underbrace{\text{curvature}\cdot\left(\ln\tfrac{K}{F}\right)^2}_{\text{wings}} \tag{11}$$

Parameter	Economic meaning	WTI typical
ATM vol σ	Base implied vol at the money. Higher for nat gas ($\sim$ 55%), lower for gold ($\sim$ 15%).	30%
Skew	Negative in most commodity markets. Reflects downside fear. More negative for crude than for gold.	-0.03
Curvature	Symmetric wing steepness. Positive values create a U-shaped smile.	$+0.05$
Steepness ν	Term structure slope. Short-dated vol > long-dated vol due to mean-reversion of commodity prices.	$+0.08$

10 Forward Curve Data Layer

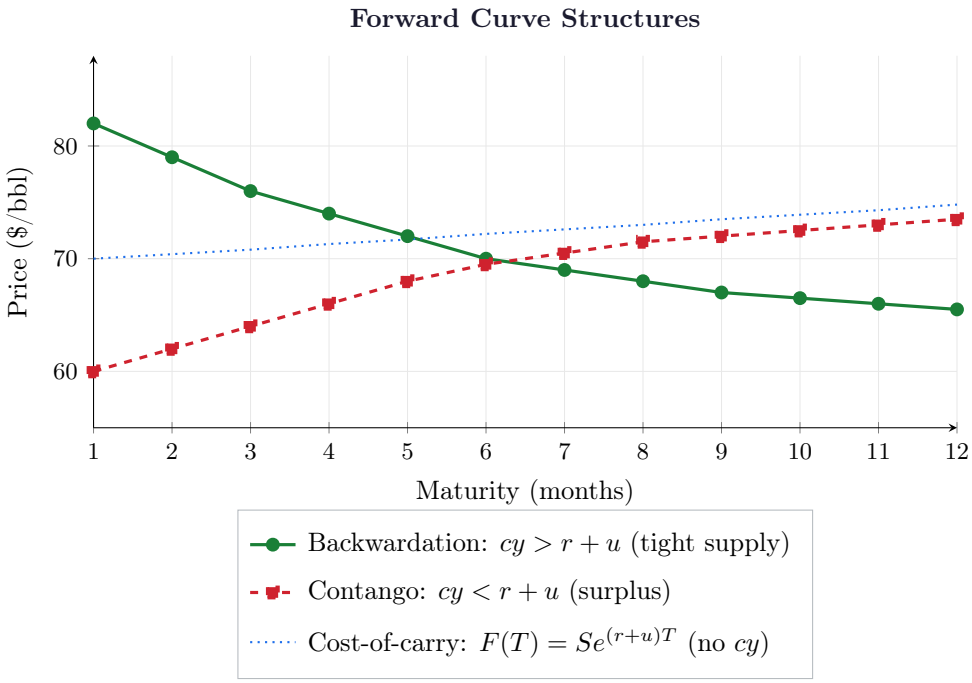
10.1 Cost-of-Carry Model

When live data is unavailable, CODAP generates a synthetic fallback curve:

$$F(T) = S \cdot e^{(r+u-cy)\,T} \tag{12}$$

where u is the annualised storage cost and cy the convenience yield.

- $cy > r + u$: **backwardation** — physical scarcity premium
- $cy < r + u$: **contango** — storage surplus, carry dominates
- $cy = r + u$: flat curve at fair value



10.2 Data Sources and Commodity Registry

Family	Count	Source	Examples
Energy	7	Yahoo Finance	WTI, Brent, Nat Gas, RBOB, Gasoil
Metals	5	Yahoo Finance	Gold, Silver, Copper, Platinum, Palladium
Agriculture	15	Yahoo Finance	Corn, Wheat, Soybeans, Sugar, Coffee, Cattle
Base Metals (LME)	9	TradingView	Copper, Aluminum, Zinc, Nickel, Lead
Energy (additional)	8	TradingView	TTF, NBP, Coal API2/4, Uranium, Carbon EUA
Freight	4	TradingView	Capesize, Panamax, Supramax, VLCC
Carbon	4	TradingView	EU EUA, UK UKA, California CCA, RGGI
Total	**65+**		

11 Applications and Extensions

11.1 Typical Users

User	How CODAP helps
Options Desk	Price vanilla and barrier options on any of 65+ commodities. Read off full Greeks for hedging. Assess the vol surface to identify whether implied vol is rich or cheap relative to ATM.
Physical Hedger	Price Asian calls based on monthly average procurement costs. Lock in a fixed price for 12+ months with a commodity swap. Compare vanilla vs barrier premium to find the optimal hedge structure.
Margin Hedger	Price 3-2-1, jet crack and spark spread options to protect gross refining or generation margins. Compute break-even spread levels.
Spread Trader	Price calendar spread options to speculate on steepening or flattening. Trade the roll yield in backwardation markets.

11.2 Possible Quantitative Extensions

Implied vol from real option chains. Yahoo Finance exposes option chain data for WTI, Gold and Corn via `yfinance`. CODAP could extract market-implied vols per strike and maturity to overlay against the parametric model.

SABR stochastic volatility. The SABR model is the industry standard for capturing the vol smile in commodity and interest rate derivatives. It produces more realistic OTM prices and is consistent with the market's implied vol surface.

Swing options. Standard in European gas and power markets (Engie, TotalEnergies, EDF). The holder has the right to take or deliver gas at a variable daily volume within contractual bounds. Pricing requires dynamic programming.

Schwartz-Smith Monte Carlo. Using calibrated SS parameters $(\kappa, \xi_0, \sigma_\chi, \sigma_\xi)$ as diffusion inputs to price path-dependent exotics in a single consistent framework.

12 Conclusion

CODAP demonstrates how a professional-grade commodity derivatives pricer can be built entirely in Python with freely available market data.

Six pricing engines cover the full spectrum of products used by commodity desks: vanilla options (Black-76), average-price options (Monte Carlo), spread options (Kirk 1995), curve slope options, fixed-for-floating swaps, and path-dependent barrier options. The expiry-aware ticker logic ensures prices always come from live, non-expired contracts.

Together with CFCAP, CODAP provides a complete analytical stack: forward curve analysis and trading signals on one side, derivatives pricing and risk on the other.

Built with Python · Yahoo Finance · TradingView